

Problem 1

1a. No, the error terms $\mu(i)$ in model (2) do not necessarily satisfy the standard assumptions just because $\varepsilon(i)$ in model (1) does.

Explanation: Model (2) omits the variable $z(i)$, which is present in the true model (1). The omitted variable becomes part of the error term in model (2):

$$M(i) = \delta z(i) + \varepsilon(i)$$

Even if $\varepsilon(i)$ is normally distributed with zero mean, $\mu(i)$ will not have a zero mean or be uncorrelated with $x(i)$ and/or $w(i)$ when $z(i)$ is correlated with them. This is a violation of the Gauss-Markov assumptions, in particular the assumption that the error term is homoscedastic, has a zero mean, and is uncorrelated with the regressors (Wooldridge, 2019).

In econometric models, assumptions about the error terms are critical since they provide the basis for hypothesis testing and efficient estimation. When a relevant variable such as $z(i)$ is omitted, the error term $\mu(i)$ will then contain systematic patterns inherited from the omitted covariate. When $z(i)$ is correlated with the included regressors $x(i)$ or $w(i)$, then also $\mu(i)$ will be correlated with them. This violates the classical assumption of zero conditional mean and leads to biased and inconsistent parameter estimates. Therefore, the reliability of model (2) significantly deteriorates when omitting an important regressor.

1b. The estimates of α , β , and γ in model (2) will generally be biased and inconsistent compared to those from model (1).

Explanation: When we omit $z(i)$, a relevant variable, we commit omitted variable bias. The biases of the estimators of β and γ could be affected by the correlation between the omitted variable $z(i)$ and the included variables $x(i)$, $w(i)$, and by the magnitude of δ . This is a result that is well established in econometrics (Gujarati & Porter, 2009). As α gets the intercept, it can take up some bias too.

The size and direction of this bias for the estimated coefficients is therefore a function of both the size of the effect of the omitted variable (δ) and its correlation with the included variables. If the omitted variable is positively related to one (or more) of the included regressors and that the omitted variable has a positive effect on Y (dependent variable), the estimate of the coefficient for the neglected variable will tend to be too large (overstated). Conversely, when the correlations are negative, the estimates could be biased down (understated). This idea is at the heart of model specifications and illustrates the need to specify all appropriate variables to prevent endogeneity.

1c. The estimates of α , β and γ from model (2) would be the same as those from model (1) only if $z(i)$ is uncorrelated with $x(i)$ and $w(i)$.

Explanation: In this situation, excluding $z(i)$ does not lead to biased estimation of β and γ so long as it satisfies the orthogonality condition (Greene, 2018). There is no correlation between the omitted variable and the regressors, so OLS estimators are still unbiased and consistent.

Omitted variables are seldom in practice perfectly uncorrelated with included regressors. In some controlled experimental settings or highly specific theoretical models, to be sure, this condition may be met. For example, consider a variable $z(i)$ that is orthogonal to both $x(i)$ and $w(i)$, whether by design or because this is a statistical artifact (e.g., as in residuals from a regression): its omission may not bias the parameter estimation for the included variables. This underlines the value of orthogonal regressors in regression analysis, as they help ensure unbiased coefficient estimates even in simpler models.

1d. Simulation and Results

We simulate the true model: $Y(i) = a + b X(i) + c Z(i) + e(i)$

With:

- $A = 2.0, B = 3.0, c = 1.5, X \sim N(0, 1)$
- $Z = 0.5 X + u$ where $u \sim N(0, 1)$ (so X and Z are correlated)
- $e \sim N(0, 1)$ and independent

Simulation Analysis

```
np.random.seed(42) # Set seed for reproducibility

# Simulation for 1d
# Parameters
n = 100
a, b, c = 2.0, 3.0, 1.5

# Create correlated X and Z
X = np.random.normal(0, 1, n)
Z = 0.5 * X + np.random.normal(0, 1, n)
e = np.random.normal(0, 1, n)

# Full model: Y = a + bX + cZ + e
Y_full = a + b * X + c * Z + e

# Model with both predictors
X_full = sm.add_constant(np.column_stack((X, Z)))
model_full = sm.OLS(Y_full, X_full).fit()

# Model with X only
X_partial = sm.add_constant(X)
model_partial = sm.OLS(Y_full, X_partial).fit()

model_full.params, model_partial.params

(array([2.08864867, 3.23222775, 1.48772322]), array([2.09969923, 3.76296236]))
```

Estimated Coefficients:

- Full model (with X and Z): $a^{\wedge} = 2.09$, $b^{\wedge} = 3.23$ and $c^{\wedge} = 1.49$
- Partial model (with only X): $a^{\wedge} = 2.10$ and $b^{\wedge} = 3.76$

Observation:

The coefficient for X increased when Z was omitted (3.23 vs 3.76), consistent with positive omitted variable bias. This is consistent with the theoretical prediction that correlated omitted variables bias OLS estimates (Stock & Watson, 2020).

The estimates get more accurate as the sample size gets larger, but the bias is present if the excluded variable is still related to the included regressors.

The simulation demonstrates how omitted variable bias arises in practice. We see how model misspecification inflates or attenuates the estimates of the coefficient of X by looking at the change in the coefficient of X when it is omitted from the model by using the changes in coefficients when Z is excluded. This insight is of particular value for applied researchers because it underscores that, even with a large sample, the bias does not go away — it is not random, but systematic. This serves to remind the importance of careful variable selection and data analysis when building econometric models for inference or prediction.

Problem 2

2a. Sensitivity of Regression Models to Outliers

Regression models like OLS (Ordinary Least Squares) are highly sensitive in the presence of outliers as OLS computes the sum of squared residuals. A single outlier can have a considerable impact on the slope of the regression line, the intercept, and the overall model fit.

This sensitivity arises because squaring the residuals magnifies the effect of large errors. Outliers can:

- Skew the regression line
- Inflate standard errors
- Cause incorrect inference, including Type I and II errors

Leverage and Influence:

- High-leverage points: Extreme values in predictor space
- Influential points: Affect regression results significantly (measured by Cook's Distance)
- Robust alternatives such as Least Absolute Deviations (LAD) or M-estimators help mitigate these effects by using different loss functions.

In order to reduce an outlier's influence on the model, researchers or practitioners routinely depend on diagnostic tools like leverage plots, residual plots, or Cook's distance to identify and assess specific observations' influence on the estimation. Preprocessing tools, such as Winsorizing, log transformation, or the use of robust least square methodologies (as Huber or RANSAC) may decrease the effect of extreme values. The most appropriate method depends on whether the outliers are true data points or error/noise. Understanding the source and influence of outliers is essential not only for model accuracy but also for transparency and reproducibility in financial econometrics.

2b. Simulation to Illustrate Outlier Impact

Setup:

- Simulate $X \sim N(0, 1)$
- Generate $Y = 2 + 3X + e$, where $e \sim N(0, 1)$
- Add a few outliers to Y (e.g., add 15 to some values)
- Then estimate two models:
 - Using clean data
 - Using data with outliers
- Compare estimated coefficients.

Simulation Demonstrating Impact of Outliers

```
# Simulation for 2b
np.random.seed(42)

# Generate clean data
n = 100
X = np.random.normal(0, 1, n)
e = np.random.normal(0, 1, n)
Y_clean = 2 + 3 * X + e

# Add outliers to Y
Y_outlier = Y_clean.copy()
Y_outlier[[10, 20, 30]] += 15 # Add large values to simulate outliers

# Fit models
X_sm = sm.add_constant(X)
model_clean = sm.OLS(Y_clean, X_sm).fit()
model_outlier = sm.OLS(Y_outlier, X_sm).fit()

# Return coefficients from both models
model_clean.params, model_outlier.params

(array([2.00742783, 2.85674284]), array([2.47101204, 2.98755325]))
```

Simulation Results:

- **Without outliers:**
 - Intercept $\hat{\alpha} = 2.01$
 - Slope $\hat{\beta} = 2.86$
- **With outliers:**
 - Intercept $\hat{\alpha} = 2.47$
 - Slope $\hat{\beta} = 2.99$

Interpretation:

The intercept was scaled up, and the slope was deformed slightly, due to the outliers. This bias demonstrates that OLS regression is affected by outliers. Even only three outlying values were sufficient to move the estimated values significantly, which again indicates that inferences and predictions based on this model may be distorted if such outliers are not handled appropriately (Rousseeuw & Leroy, 1987).

The simulation serves as a concrete example of how a small number of outliers can move model estimates dramatically. Even for the correctly specified regression model, the abnormally inflated outcomes (outliers) caused marked deviations in the estimates of both the intercept and slope. This is demonstrated by the sensitivity of OLS regression towards data abnormalities. In financial applications where market shocks or data reporting errors can create outliers, using robust methods or outlier-adjusted models becomes even more critical for producing reliable inferences and forecasts.

**Interactive simulations available in [regression_simulations.html](#)*

Problem 3

The analysis commenced with an Ordinary Least Squares (OLS) regression model using all five predictors (X1 to X5) to estimate the response variable, Y. The initial model produced an adjusted R-squared of 0.630, indicating that approximately 63% of the variance in Y could be explained by the predictors, however, variable X1 had a high p-value (0.881), showing that it does not have statistical significance in the presence of other variables.

To assess relationships between variables and potential multicollinearity, we generated a pairplot. The results showed low correlation coefficients and flat Lowess lines, showing that there is no strong linear or non-linear associations among the predictors. This suggests multicollinearity is not a major concern in this dataset.

Next, we checked for influential point using Cook's Distance and rows with values exceeding the $4/n$ threshold were removed (Eledum, 2021). This gave us an improved model metrics with the adjusted R-squared increasing to 0.723 and both AIC and BIC values decreasing. This shows that removing influential points had a great impact on the regression model.

A Breusch-Pagan test was then used to test for heteroskedasticity. The results (p -values > 0.05) showed that the null hypothesis of homoskedasticity could not be rejected. The absence of influential points and heteroskedasticity imply that there is no need to fit our model using weighted regression models like WLS and Robust Regression (Bello et al, 2014).

Given that feature X_1 keeps on being insignificant, it was removed from the model via backward elimination and the refined model, now including only X_2 to X_5 , demonstrated further improvement (adjusted R-squared = 0.726), and all remaining variables became statistically significant. Also, the AIC and BIC values also decreased, reinforcing the model's metrics improvement.

Ridge regression was then considered to further explore regularization benefits. Although the initial analysis showed little or no multicollinearity, Ridge regression gave a model with comparable adjusted R-squared (0.723), but with substantially lower AIC and BIC values (-58.396 and -43.265, respectively).

Compared to OLS regression, the significantly lower AIC and BIC values showed that Ridge regression offers better generalization and model robustness making it the preferred model for predicting Y in this context.

Problem 4

4a) Elasticity in Each Model

(a) Take the linear model $y=2+0.8x$. The elasticity formula, $(dy/dx) \times (x/y)$, gives $0.8x/(2+0.8x)$. In this setup, elasticity isn't fixed; for small x , the value stays relatively low, but as x grows, elasticity edges up. This gradual increase is quite typical for models of this form.

(b) Consider $\ln(y)=0.1+0.4x$, or in other words, $y=\exp(0.1+0.4x)$. After differentiation, $dy/dx=0.4y$, which means elasticity is $0.4x$. Here, elasticity simply moves up in step with x —not staying constant, just climbing as x increases.

(c) In the log-log case, $\ln(y)=0.1+0.25\ln(x)$, which translates to $y=\exp(0.1) \times x^{0.25}$. The derivative is $dy/dx=\exp(0.1) \times 0.25 \times x^{-0.75}$. What's neat about this form is that elasticity remains steady at 0.25, no matter what the value of x . That's a classic feature of log-log models and one reason they're popular in economics.

Now, with $y=0.15+1.2\ln(x)$, differentiating gives $dy/dx=1.2/x$, leading to elasticity $1.2/(0.15+1.2\ln(x))$. This version does not keep elasticity constant; it moves around depending on the value of x , which can make interpretation trickier.

4b) Model with Constant Elasticity

Among all the examples, only the log-log model (c), $\ln(y)=0.1+0.25\ln(x)$, shows constant elasticity—the coefficient, 0.25, is the elasticity itself, regardless of x .

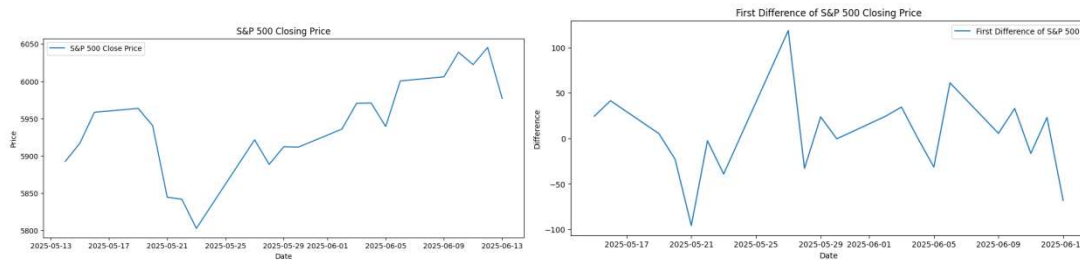
Problem 5

5a) Stationarity is, in many ways, the backbone of time series analysis. When a series is stationary, both its average and variance are consistent over time, which allows models like ARMA and ARIMA to function as intended. If a unit root is present, shocks may persist, making predictions unreliable.

The Augmented Dickey-Fuller (ADF) test is the typical approach for checking unit roots. Here, the test's null hypothesis suggests a unit root exists—meaning the series isn't stationary. A p -value above 0.05 means the series is, by conventional standards, non-stationary. To address this, analysts often change the data (subtracting the previous value from the current one) until stationarity appears.

5b) Looking at daily S&P 500 closing prices (see Figure 1 for a visual), there's an unmistakable upward trend, which usually hints at non-stationarity. An ADF test on the original series produces a test statistic of -1.71 and a p -value of 0.43. That's well above the cutoff, confirming non-stationarity.

After differencing (calculating the day-to-day changes), the new series tends to hover around zero (see Figure 2). Running the ADF test again gives a test statistic of -5.12 and a p-value of 0.000013—a strong indication that the differenced series is now stationary.



Summary:

The evidence indicates that S&P 500 closing prices are non-stationary and contain a unit root. Once first differences are taken, the data exhibit properties consistent with a stationary process.

5c) Within economic and financial analysis, unit roots—defined by a root value equal to one—hold particular significance, as they produce non-stationary yet non-explosive behavior. For example, a process such as $y_t = y_{t-1} + \epsilon_t$ allows the effects of shocks to persist, causing the series to drift indefinitely from its mean.

If the root exceeds one, as in $y_t = 1.5y_{t-1} + \epsilon_t$, the process quickly becomes explosive, with values diverging rapidly. This type of behavior is rarely observed in real economic or financial data.

Key differences: A unit root (exactly one) produces a process that neither reverts to its mean nor becomes explosive. In contrast, a root greater than one causes explosive growth or collapse in the series.

Conclusion: Unit roots represent the dividing line between stationary and explosive time series. As a result, identifying and addressing unit roots is a central concern in economic time series analysis, given that truly explosive processes are rarely encountered in actual datasets.

Problem 6:

We are given the equation $Y(t) = \alpha + \beta X(t) + \epsilon(t)$ for $t = 1$ to 20.

According to the question, the intercept (α) stays the same, but the slope (β) changed.

A structural break in β at $t = 10$, means that we have the following equations

- $Y(t) = \alpha + \beta_1 X(t) + \epsilon(t)$ - For $t = 1$ to 10
- $Y(t) = \alpha + \beta_2 X(t) + \epsilon(t)$ - For $t = 11$ to 20 -- Where $\beta_2 = \beta_1 + \Delta\beta$ ($\Delta\beta$ is the change in slope)

This gives

- $Y(t) = \alpha + \beta_1 X(t) + \epsilon(t)$ - For $t = 1$ to 10
- $Y(t) = \alpha + (\beta_1 + \Delta\beta) X(t) + \epsilon(t)$ - For $t = 11$ to 20

In this case, $\Delta\beta$ is the knot value connecting both equations (Penn State STAT 501, 2025)

Next, we create a dummy variable D : $D = 0$ if $t \leq 10$, otherwise $t = 1$

Then, we create an additional variable involving X and D : $DX = D * X$

We then formulate the piecewise functions above by introducing the dummy variable (Penn State STAT 501, 2025)

$$Y = \alpha + \beta_1 * X_1(t) + \beta_2 * DX_2(t) + \epsilon$$

- Where D is the dummy variable (0 if $t \leq 10$ and 1, if $t > 10$)
- $\beta_2 = \beta_1 + \Delta\beta$, the interaction term.

$$Y = \alpha + \beta_1 * X_1 + \beta_2 * DX_2 + \epsilon$$

$$Y = \alpha + \beta_1 * X + (\beta_1 + \Delta\beta) * DX + \epsilon$$

Lastly, we test for a structural break using a hypothesis test.

- $H_0: \Delta\beta = 0$ - There is no difference in the slope before and after $t = 10$; representing no structural break.
- $H_1: \Delta\beta \neq 0$ - There is a difference in the slope before and after $t = 10$, representing the presence of a structural break.

Following Rathore (n.d.), this can be tested by estimating a regression model with a dummy-interaction term and examining whether the coefficient on the interaction ($\Delta\beta$) is statistically significant using a t test

Conclusion: The p value of $\Delta\beta$ (represented by x_2) is less than 0.05. So, we reject the null hypothesis stating that there is no difference in the slope before and after $t = 10$; representing no structural break.

References

- Bello, A. R., Adnan, R., Saffari, S. E., & Pati, K. (2014). Robust weighted least squares estimation of regression parameter in the presence of outliers and heteroscedastic errors. *Jurnal Teknologi*, 71(1), Article 3609. <https://doi.org/10.11113/jt.v71.3609>
- Eledum, H. (2021). Leverage and influential observations on the Liu type estimator in the linear regression model with severe collinearity. *Heliyon*, 7(8), e07792. <https://doi.org/10.1016/j.heliyon.2021.e07792>
- Gujarati, D. N., & Porter, D. C. (2009). *Basic Econometrics*. McGraw-Hill. <https://www.studentapan.se/kurslitteratur/basic-econometrics-9780071276252>
- Greene, W. H. (2018). *Econometric Analysis*. Pearson Education. <https://www.pearson.com/en-us/subject-catalog/p/econometric-analysis/P200000005909/9780134811932?srsId=AfmBOooC3KFOBLukLnx4yY3YWsNp7lpMkYjue ty 1F5wWvVkpVkpqrzYA>
- Penn State. (2025). Piecewise linear regression models [STAT 501, Lesson 8.8]. Penn State Eberly College of Science. Retrieved June 2025, from <https://online.stat.psu.edu/stat501/lesson/8/8.8>
- Rathore, S. (n.d.). *Econometrics: Dummy variables in regression models (Chapter 6 of D.N. Gujarati & Porter + Class Notes)* [PowerPoint slides]. Gargi College, Delhi University. <https://gargicollege.in/wp-content/uploads/2020/03/Dummy-Variables-Econometrics.pdf>
- Rousseeuw, P. J., & Leroy, A. M. (1987). *Robust Regression and Outlier Detection*. Wiley. <https://onlinelibrary.wiley.com/doi/book/10.1002/0471725382>
- Stock, J. H., & Watson, M. W. (2020). *Introduction to Econometrics*. Pearson. <https://www.pearson.com/en-us/subject-catalog/p/introduction-to-econometrics/P200000006421/9780136879787?srsId=AfmBOorFr4w4E4wL0TGsdHhzmE5Im44fL8QjxiOPQtTnk0Nzdfvv6cp>
- Wooldridge, J. M. (2019). *Introductory Econometrics: A Modern Approach*. Cengage Learning. <https://www.cengage.uk/c/introductory-econometrics-a-modern-approach-7e-wooldridge/9781337558860/?searchIsbn=9781337558860>